Microsoft

Power BI for Developers: PBIP and CI/CD Unleashed

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Schedule

Time	Topic	Labs
9:00 – 11:30 *10:30 – Coffee Break*	**Power BI Project (PBIP) fundamentals**	Lab 1 – PBIP fundamentals
11:30 – 12:30	**Power BI & CI/CD**	Lab 2 – Power BI & CI/CD
12:30 – 13:30	*Lunch*	
13:30 – 14:30	**Power BI & CI/CD - Continue**	
14:30 – 16:30 *15:00 – Coffee Break*	**Power BI Development with AI**	Lab 3 - Power BI Development with AI
16:30 – 17:00	**Open discussion / Q&A**	

Requirements

- Licenses
  - Access to Fabric tenant [we can provide]
  - GitHub Account
  - GitHub Copilot Pro License [we can provide]
  - Access to create and authorize a service principal [we can provide]

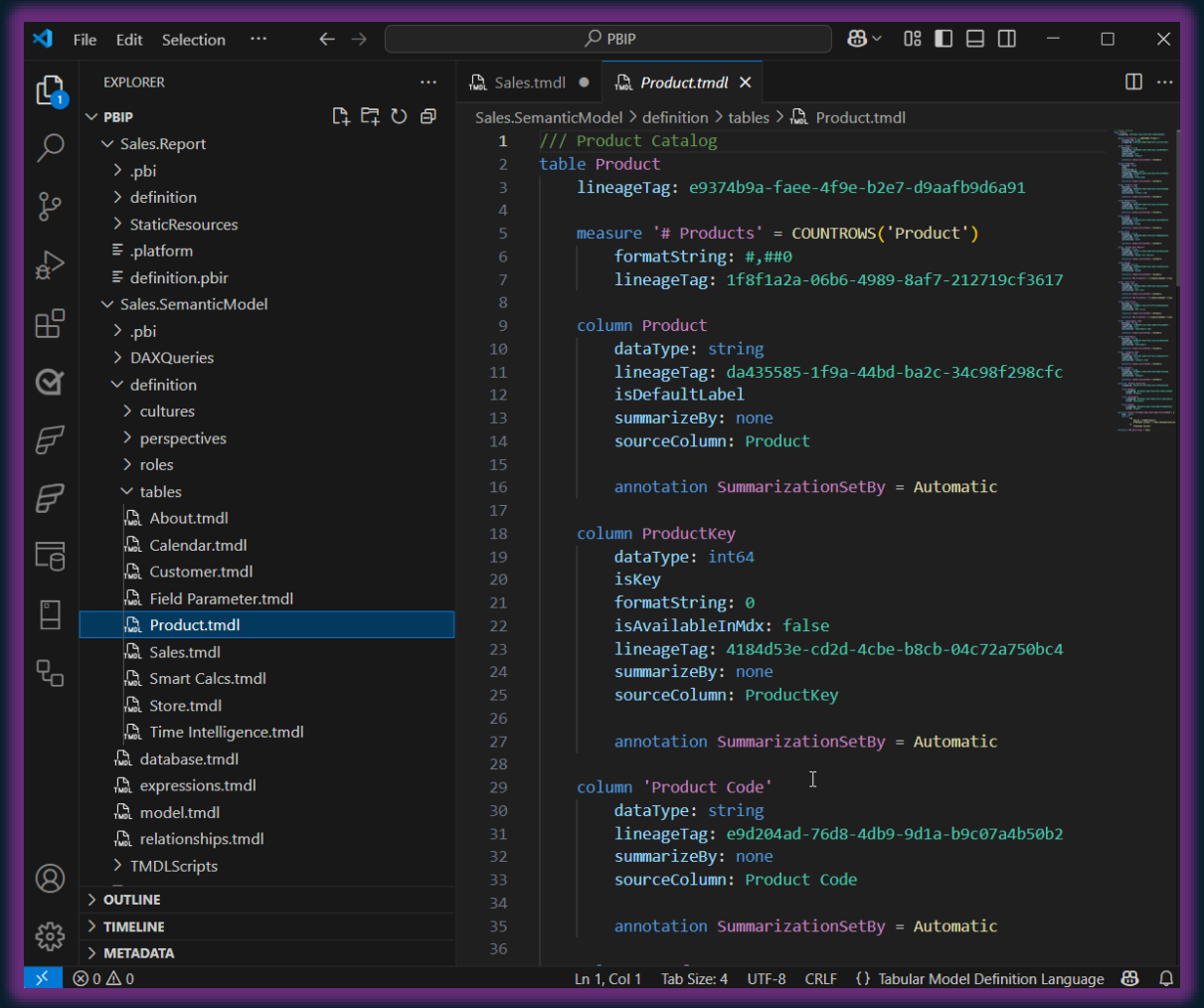
- Software
  - Power BI Desktop February 2026 release
  - Visual Studio Code
  - Git for Windows
  - Python 3.12

Workshop resources

https://tinyurl.com/workshop-pbig-2026

Use this form if you need accounts/licenses for the workshop
* WIFI: PBIG26-Participant – YourD@taJourney+Together-PBIG2026

Power BI Development in the Age of AI



{ } Code

Open file formats.

🤝 Collaborate

Work together to achieve more.

⚡ Accelerate

Efficiency boost with AI: 10x > E2E

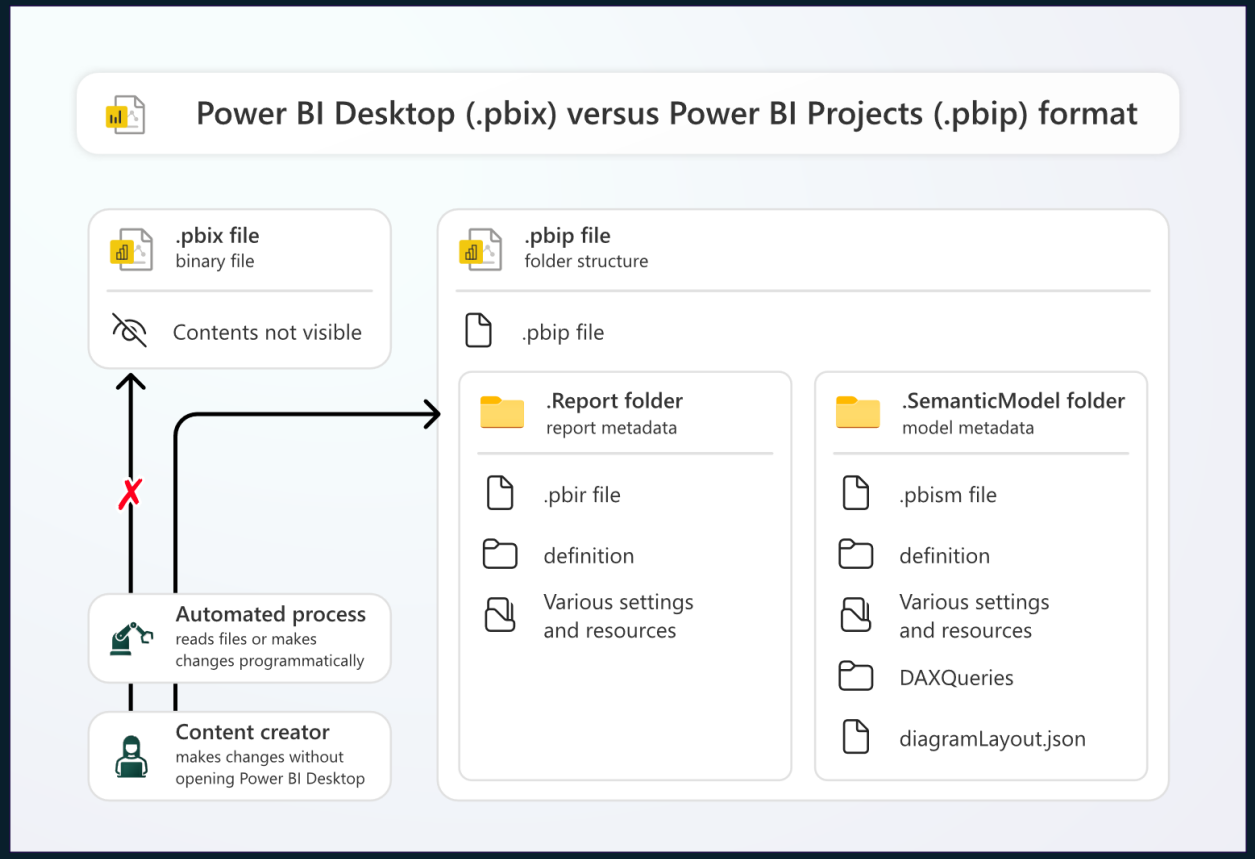
Power BI Project (PBIP)

Open, documented code formats for reports (PBIR) and semantic models (TMDL).

Code-behind

Supports code changes

Unblocks collaboration and powerful CI/CD flows



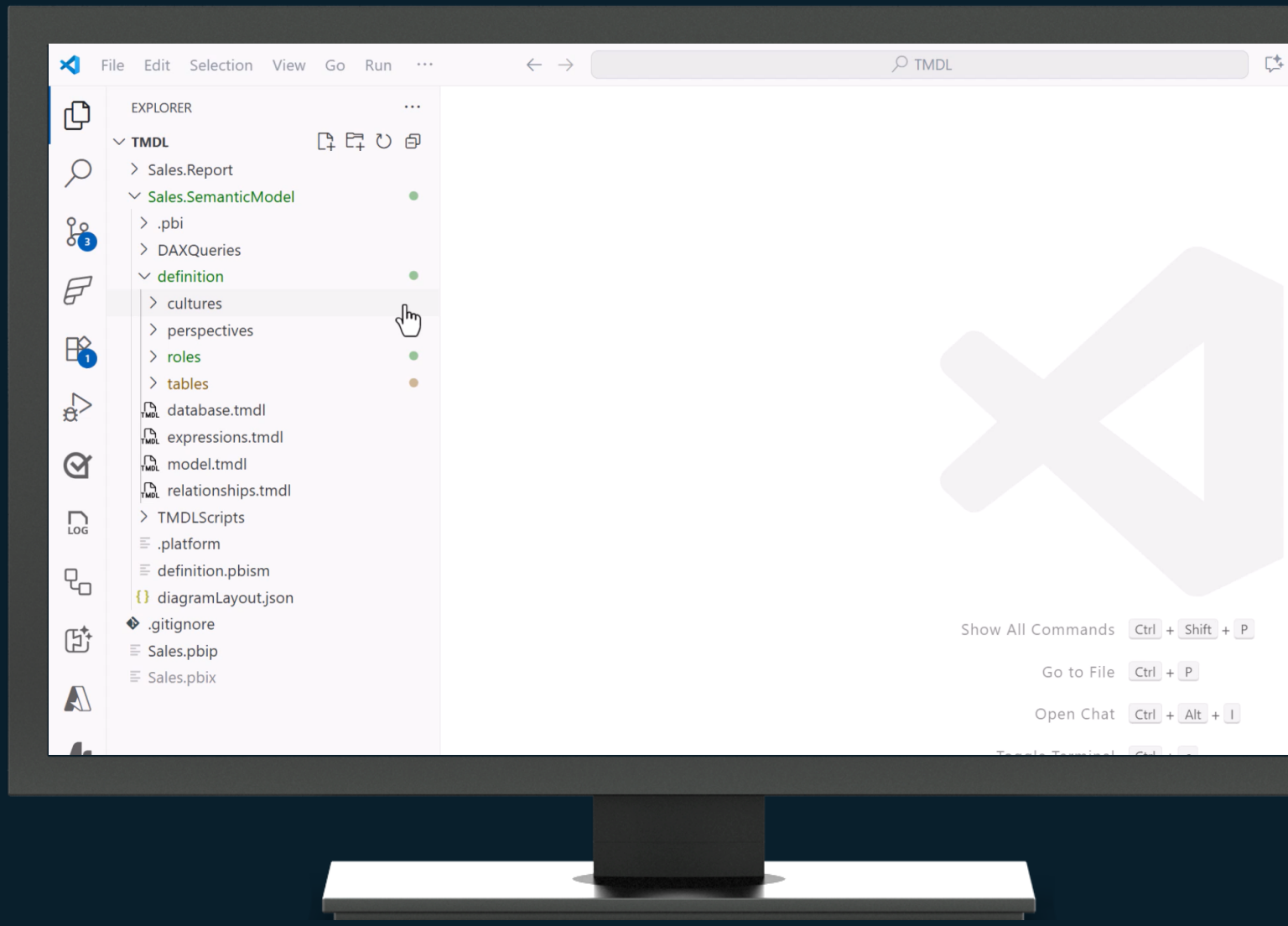
GA planned for Q3 2026

Tabular Model Definition Language (TMDL)

📄 **Declarative & Readable**
Define semantic models using clean, human-friendly text files

♻️ **Modular & Reusable**
Reuse model components across projects to drive consistency and efficiency

🤝 **Built for Collaboration**
Track changes, review diffs, and manage versions with Git



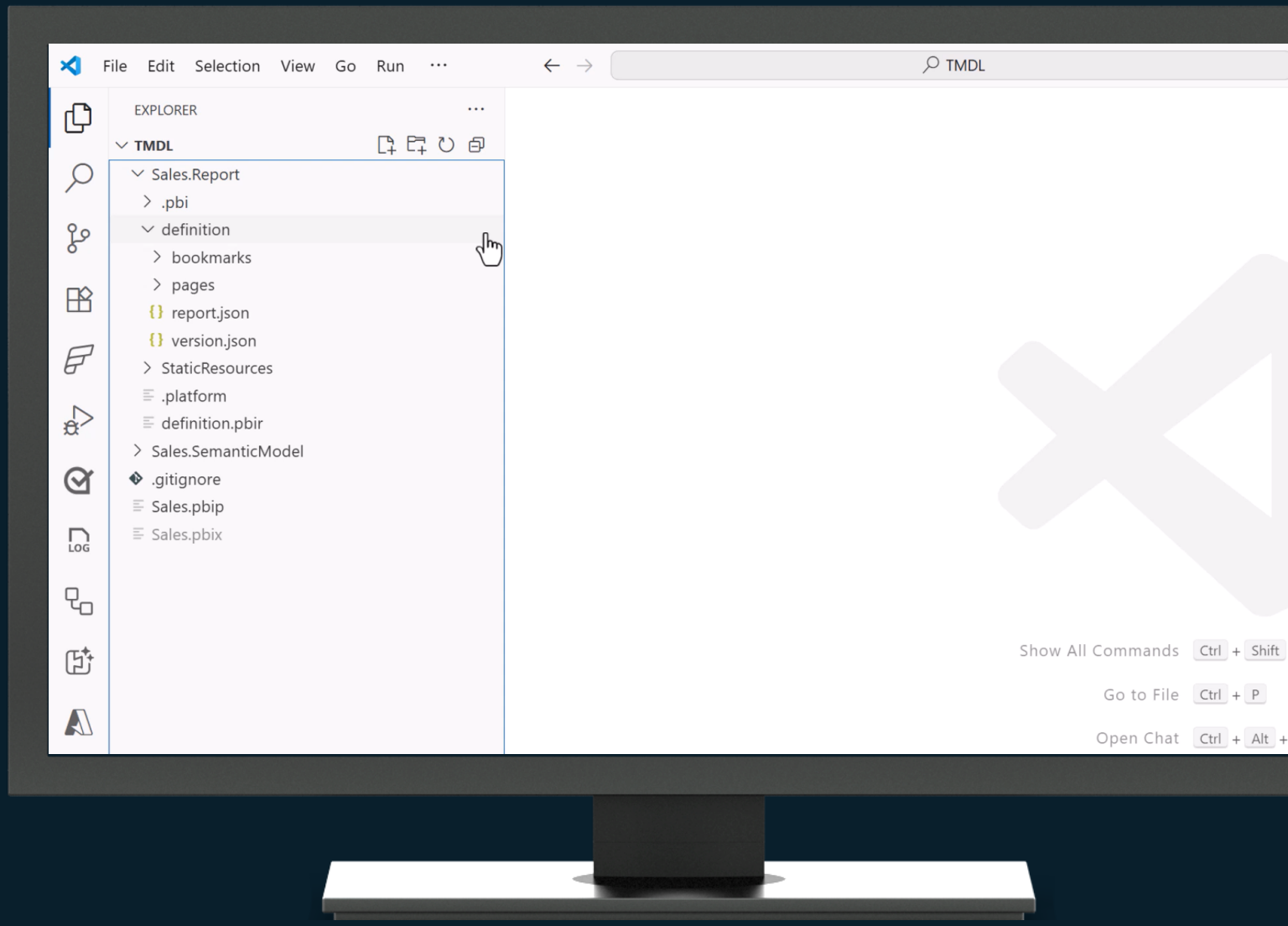
https://aka.ms/pbidevmode-tmdl

Power **BI** enhanced **R**eport format (**PBIR**)

🌐 **Public & Documented**
Thoroughly documented with up-to-date JSON Schemas

📄 **Developer-Ready**
Source-control-friendly JSON folder structure.

♻️ **Modular & Efficient**
Copy-paste visuals/pages/bookmarks/… across reports.



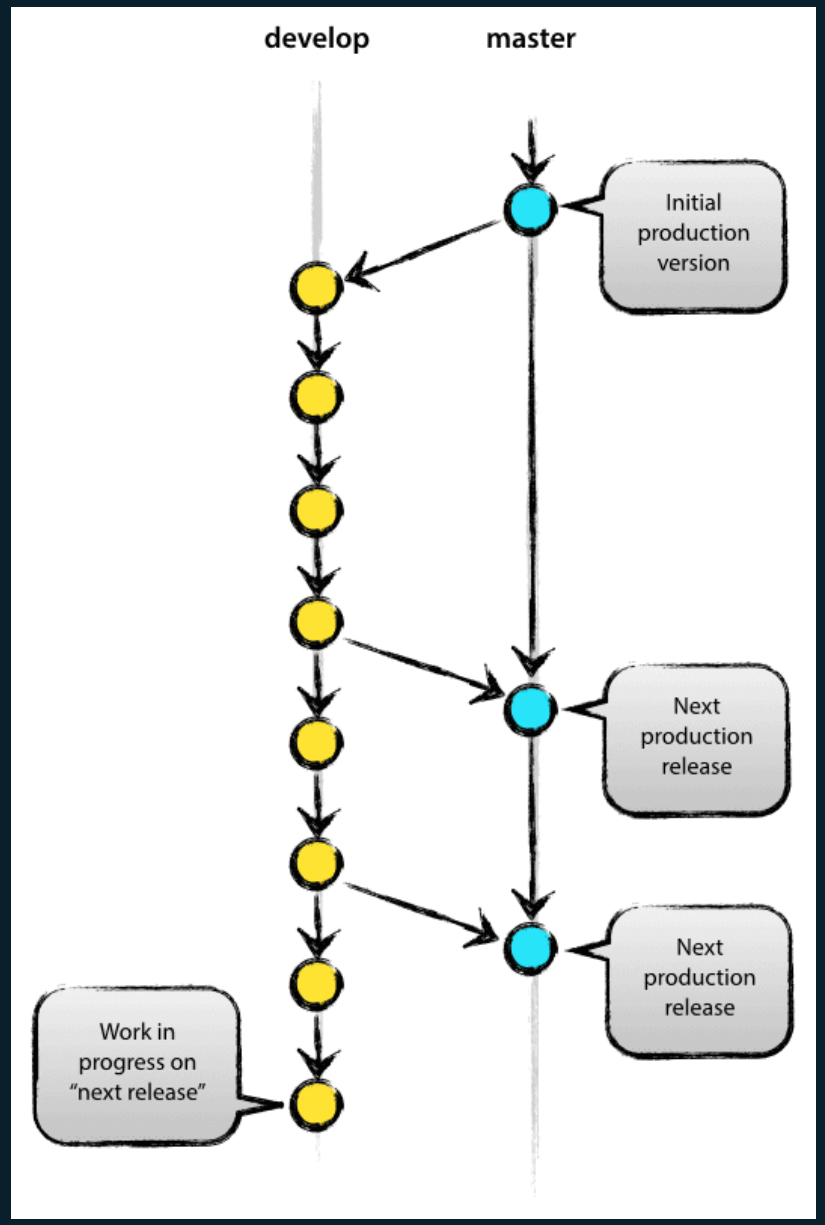
https://aka.ms/pbidevmode-pbir

Git Basics

- Git is a database (also known as repository)!
- It tracks all changes you made to your files
- With Git you don't need to store multiple physical copies of your files
- No need to start with (scary) command lines, VS Code and Copilot can help you!

Main (basic) Concepts

- **Commit:** a saved snapshot of your files at a point in time.
- **History:** A timeline of all commits, showing what changed and when.
- **Branch:** A parallel version of your work where you can experiment safely.
- **Merge:** Combining changes from one branch into another (usually into main).



LAB 1

MODERN DATA DEPLOYMENT: CI/CD FOR DATA ENGINEERS

CONTINUOUS INTEGRATION (CI)

DATA ENGINEER → COMMIT → CI → AUTOMATED VALIDATION (e.g., Unit & Schema Tests)

Automatically validating and testing code changes.

CONTINUOUS DELIVERY & DEPLOYMENT (CD)

VALIDATED CHANGES → DEPLOY TO STAGING → APPROVAL → DEPLOY TO PRODUCTION

Automating deployment to staging and production environments.

1. CODE REPOSITORY

Code, Notebooks, Configs

2. AUTOMATED TESTING

Schema & Logic Validation

3. DEPLOY TO STAGING

Verify in Test

4. DEPLOY TO PRODUCTION

Go Live with Confidence

VELOCITY

- Faster updates
- More frequent releases

RELIABILITY

- Consistent data quality
- Fewer errors in production

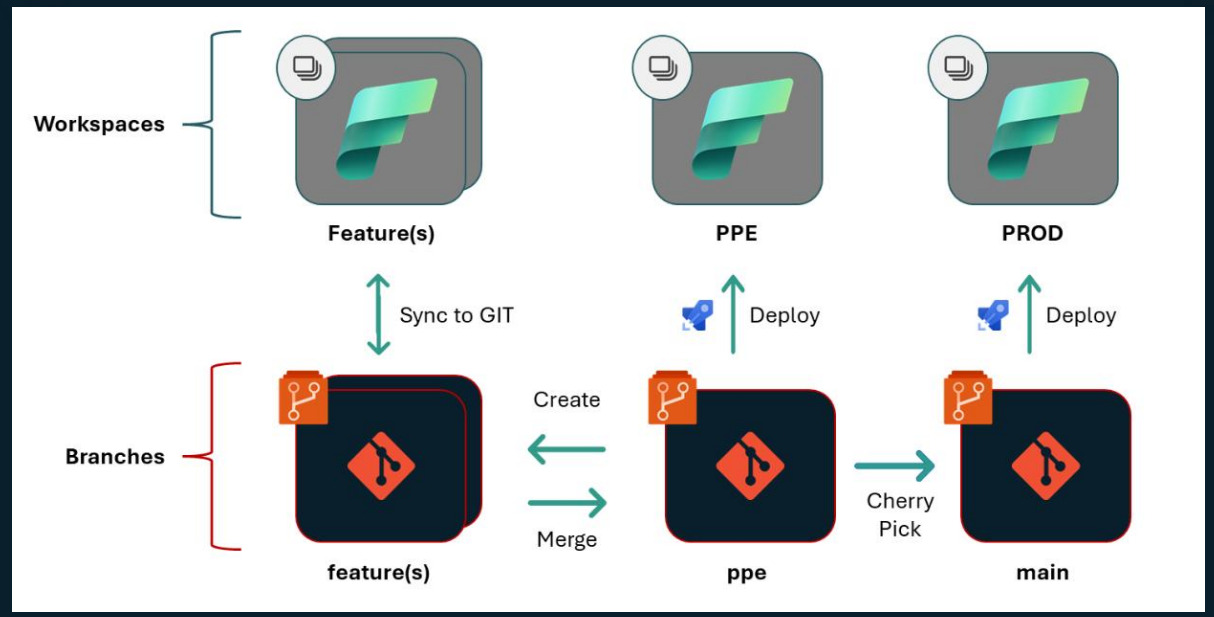
RISK REDUCTION

- Auditable deployments
- Easier rollbacks

Fabric-CICD

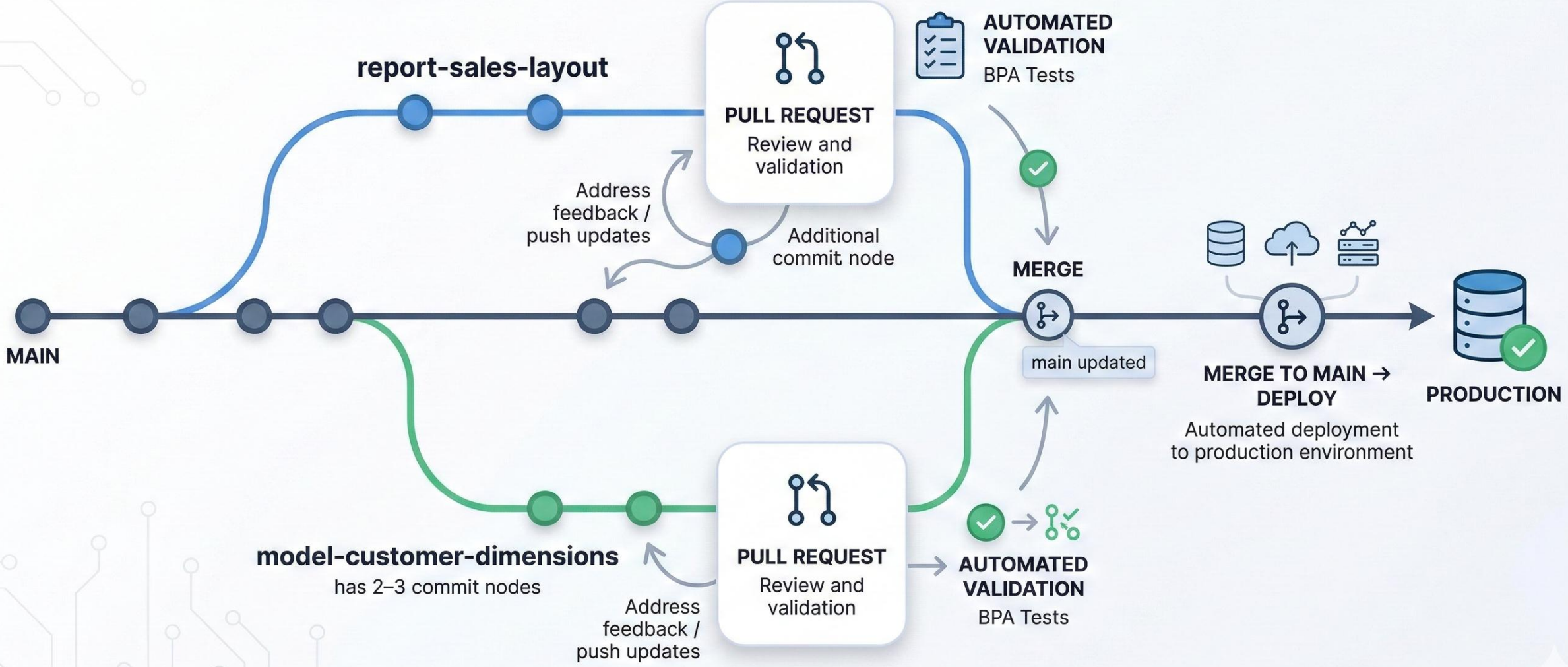
- Python library designed for Fabric workspace automation
- Supports code-first CI/CD automations to integrate Source Controlled workspaces into a deployment framework.
- Officially supported by Microsoft

```python
from fabric_cicd import FabricWorkspace, publish_all_items, unpublish_all_orphan_items

# Initialize the FabricWorkspace object with the required parameters
target_workspace = FabricWorkspace(
    workspace_id = "your-workspace-id",
    environment = "your-target-environment",
    repository_directory = "your-repository-directory",
    item_type_in_scope = ["Notebook", "DataPipeline", "Environment"],
)

# Publish all items defined in item_type_in_scope
publish_all_items(target_workspace)
```

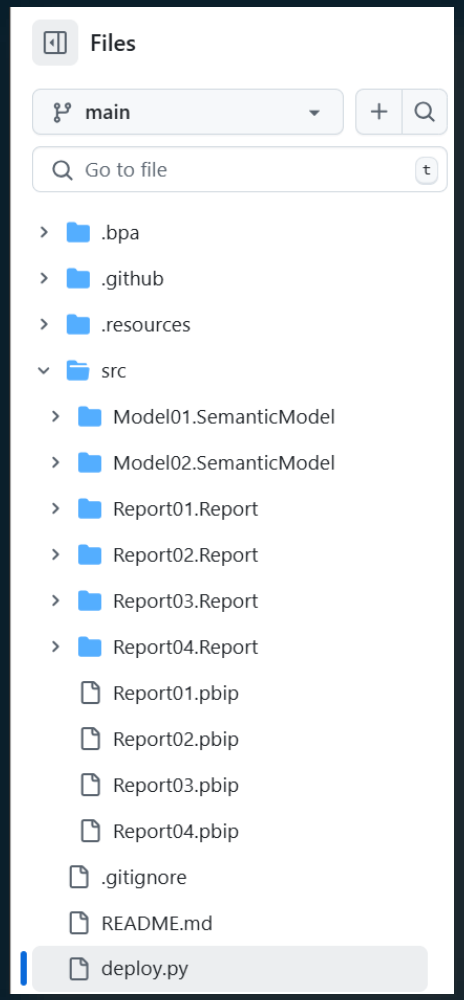


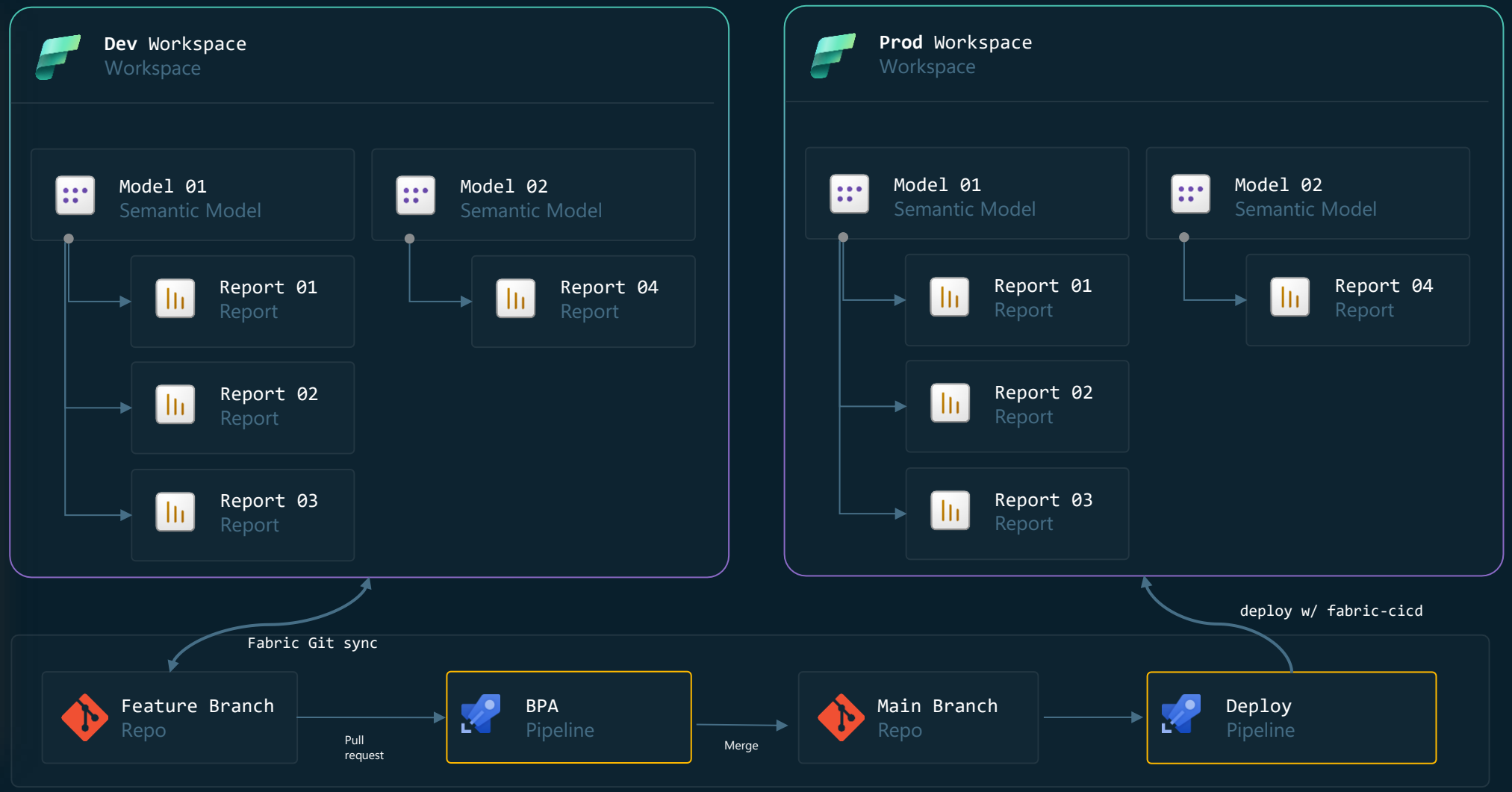
https://microsoft.github.io/fabric-cicd

SIMPLE GIT WORKFLOW FOR DATA PROJECTS



Power BI CI/CD – A possible pro-code approach





LAB 2

TMDL view – Modeling scripting experience

👀 See Everything

Dive deep into your entire semantic model metadata

⚡ Work Smarter, Faster

Make bulk edits with a powerful code editor - manually or with AI 🤖

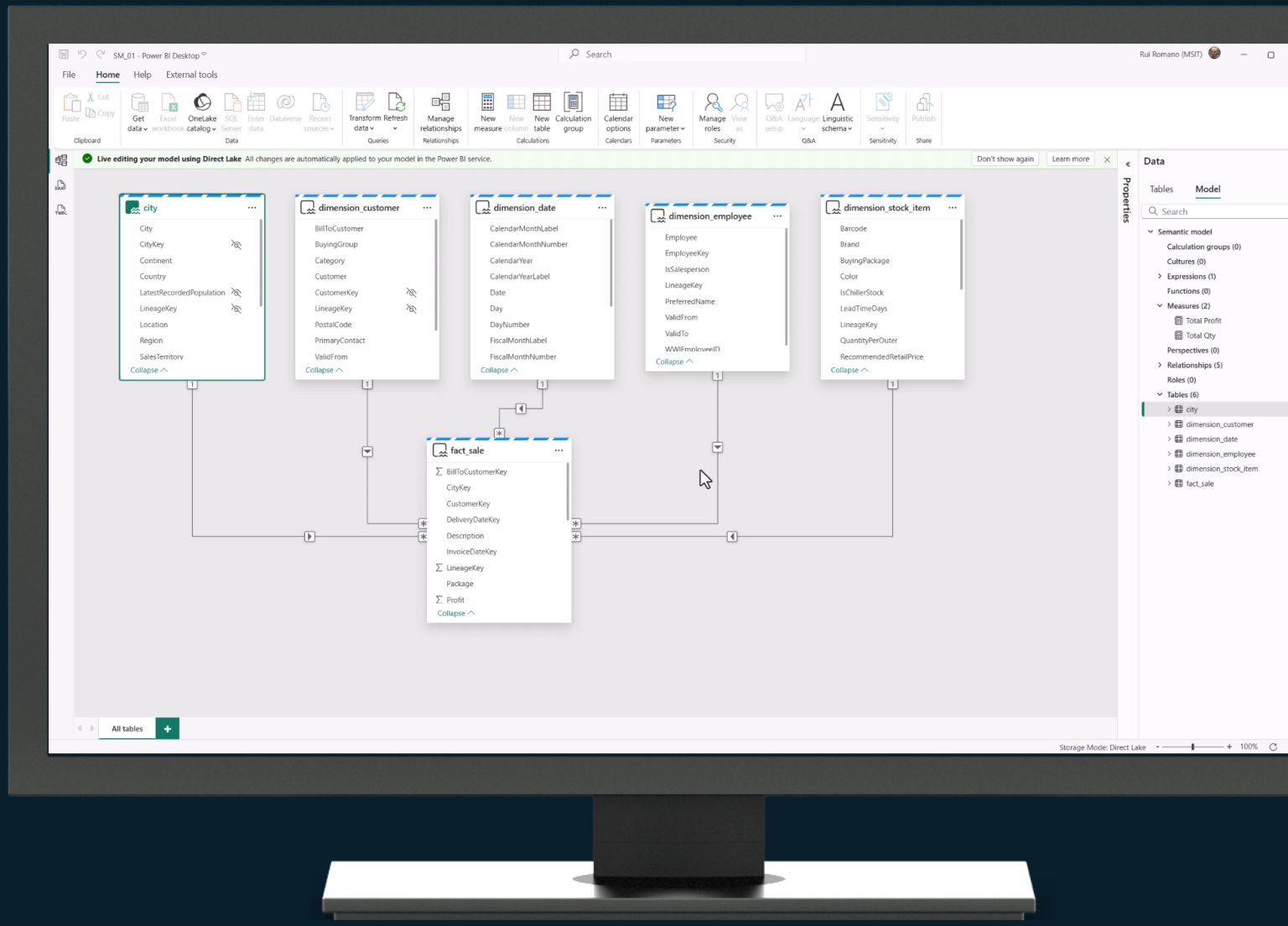
🛠️ Take Full Control

Edit advanced objects and properties not available in the UI

♻️ Reuse Everywhere

Leverage TMDL scripts to streamline modeling across multiple projects.

https://aka.ms/tmdl-gallery



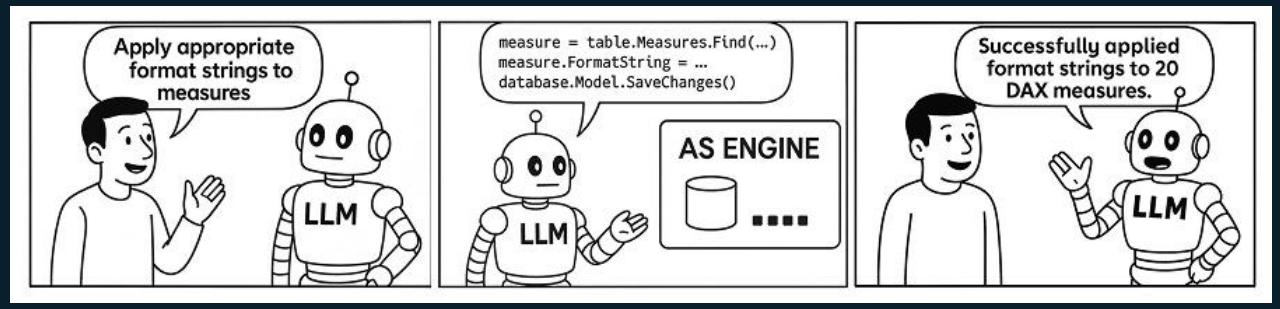
Power BI Modeling MCP Server

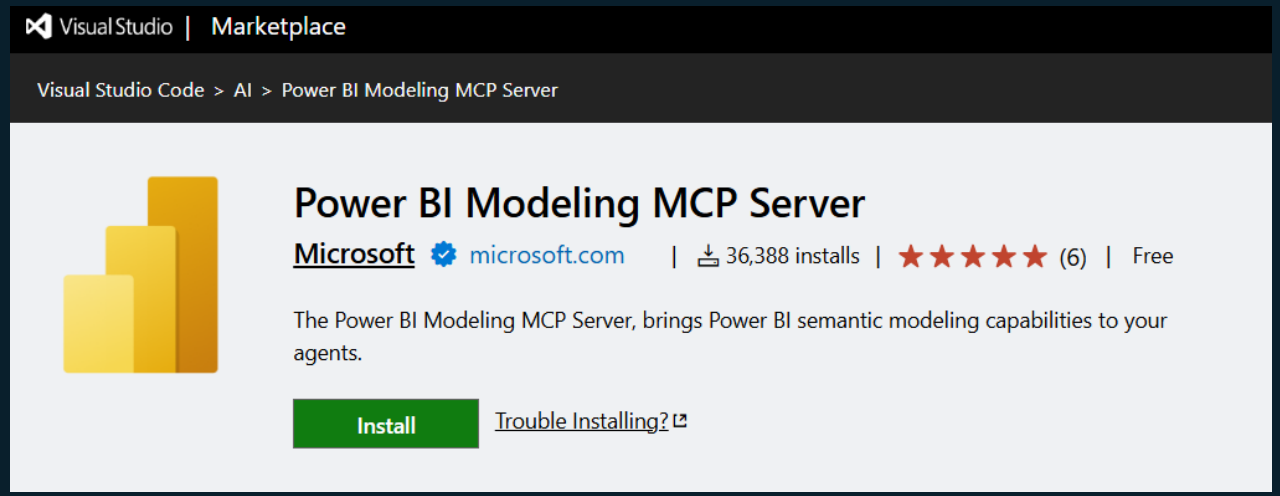
Allows AI tools to interact with Power BI's semantic layer

Local MCP Server: distributed as a VS Code extension

10x Efficiency: Easily apply bulk changes turning hours of work into seconds; enforce team best-practices and more

Agentic workflows: Together with skills and context unblock powerful E2E AI development workflows.





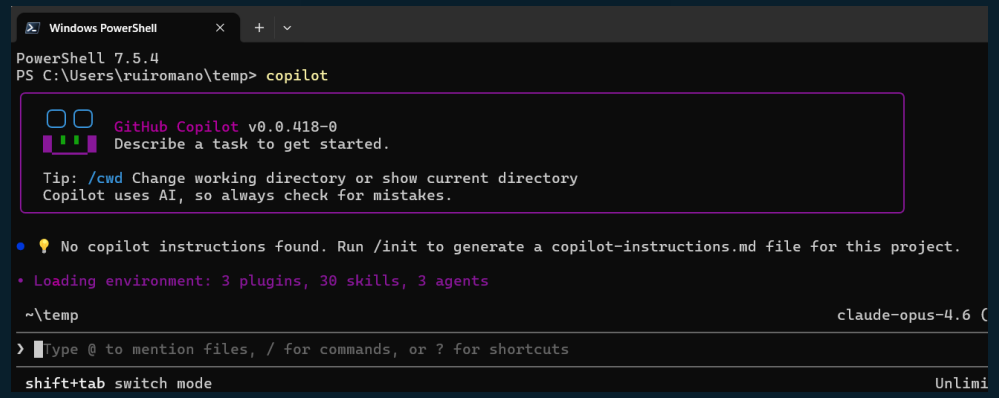
http://aka.ms/powerbi-modeling-mcp

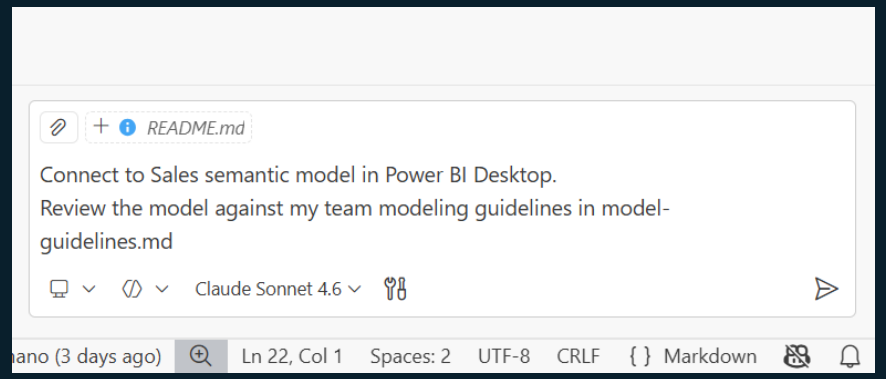
GitHub Copilot

AI partner for building solutions - not just writing code - helps developers design, create, and operate software.

Works across multiple environments - Developers can use Copilot inside Visual Studio Code, from the command line (CLI), or in cloud environments with coding agents that execute tasks remotely.

Acts as a collaborative development agent - Copilot can reason over a project, understand context, and help developers move from idea → implementation → deployment with AI assistance.





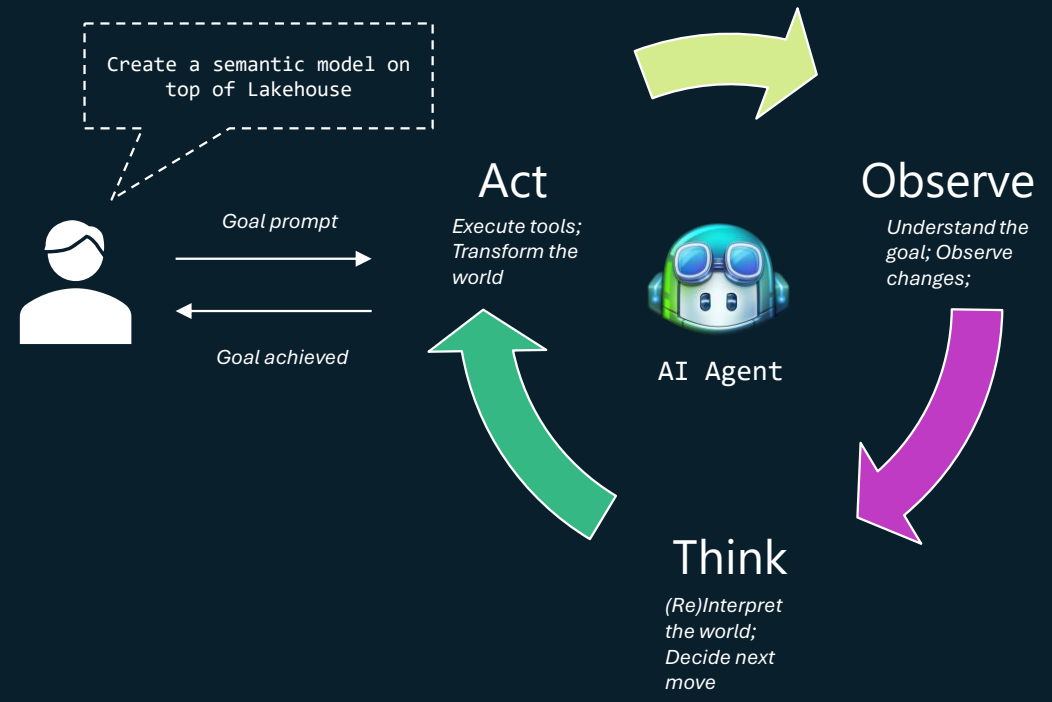
Agentic Power BI Development

Power BI developer steer, AI executes and deliver 🤖

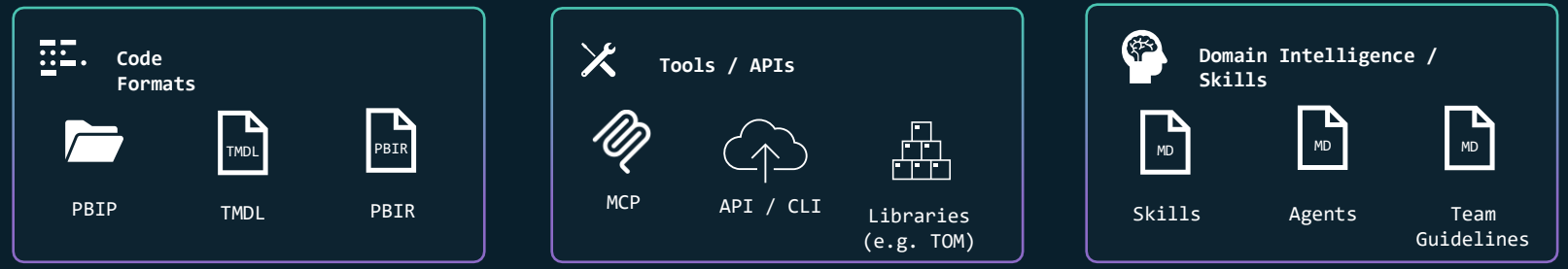
Power BI's shift to open code - enabling AI to work directly against the code with domain-specific context.

Developer's focus on defining the intent and expected outcome. "Specs are the new source code" (Sean Grove, OpenAI)

AI Agentic loop repeating cycle where AI Agent recursively prompts itself to achieve a goal learning from it's own actions.




```
Goal: Create Semantic Model

Loop 1:
-  Observe: Understand goal
-  Think: Need schema information
-  Act: Call GetSchema

Loop 2:
-  Observe: Schema received
-  Think: Design Model
-  Act: Call CreateModel

Loop 3:
-  Observe: Validation issues
-  Decide: Diagnose problems
-  Act: Call UpdateRelationships

…

Until user goal is achieved.
```



https://aka.ms/powerbi-agentic

Important Concepts

Agentic AI – Autonomous AI that plans and completes multi-step tasks
MCP Servers – Standard protocol enabling LLMs to securely access external tools and data
Custom Instructions – Configure how the agent behaves and responds
Agent Skills – Reusable capabilities that extend what an agent can do
Context Size – Maximum token window for memory and reasoning
Safety & Guardrails – Control what the agent can see and execute
Privacy Risk – Prompts, history, and tool results may be sent to external LLM providers
Context Rot – Long conversations can reduce model accuracy
Error Compounding – Small mistakes can cascade across steps

LAB 3

Loved it? Learned something? Tell us!
Share your feedback in just 1 minute

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PBIG
POWER BI GEBRUIKERSGROEP

A big thank you to our partners

plainwater – de kracht van heldere data
webdashboard – Powered by In Summa
sogeti – Part of Capgemini
DashData – power to your people
BEILO BI ZEELAND
infotopics – apps for power bi
Motio
Kimura
FabriCode
SIGNON – ICT TRAININGEN
valcon
KASPAROV FINANCE & BI
ilionx
creates.
nine altitudes
Tabular Editor – Better Data Models Faster
raedt-BI
OKVIZ
Blauwdruk BI
THE DATA COOKS
DataShareHub
easydash
DTX
GET RESPONSIVE
DataHues.
Quanto – collective analytics
ANOTHER DIMENSION
Control Over Reports
DATAKINGDOM
ONE PORTAL
DataCentral
dexs
Barbanson – Data Solutions
mountdata – guide to impact
Fellowmind
DATATAKO
AXELIO
MINOVA

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